

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

24th May 2023

Subject CM1A – Actuarial Mathematics (Paper A)

Time allowed: 3 Hours 15 Minutes (10.15 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

Q. 1) Describe the characteristics of following:

i) Index-linked security (3)

ii) Equity investment (3)
[6]

Q. 2) State the advantages and disadvantages of using models in actuarial work. [5]

Q. 3)

i) Define Ultimate mortality and Select mortality. Explain why Select mortality is lower than Ultimate mortality at a given age. (3)

ii) Given the following Select and Ultimate mortality data (i.e., q_x), calculate the values of $l_{[x]}$ and $l_{[x]+1}$ for $x = 45$ and $x = 46$, assuming that $l_{47} = 10,000$.

Age	Duration 0	Duration 1	Duration 2+ (Ultimate)
45	0.001201		
46	0.001320	0.001557	
47	0.001455	0.001735	0.001802
48		0.001938	0.002008
49			0.002241

(4)
[7]

Q. 4) An investor purchases a bond on the issue date at a price of Rs. 96 per 100 nominal. Coupons at an annual rate of 4% are paid annually in arrears. The bond will be redeemed at par 20 years after the issue date. Calculate the gross redemption yield of the bond. [3]

Q. 5) A term assurance contract for a life aged 45 years exact with a policy term of 10 years provides a benefit of Rs. 100,000 payable at the end of the year of death. Calculate the expected present value and variance of benefits payable under this contract using the basis below:

Basis:

Mortality: AM92 Select

Interest: 4% per annum

[5]

Q. 6) A bank has been accused for charging high interest rates on short-duration consumer loans, which are being provided to high-risk customers. Repayments are collected in-person by representatives of the bank making the loan. The initial loans are for Rs. 2,000 and repayments are made at an annual rate of Rs. 2,400 payable monthly in advance for two years.

The consumer forum says that “on this particular type of loan, consumers who make all their repayments pay interest at an annual effective rate of over 200%”.

The bank states that “on the same loans, 40% of the consumers default on all their remaining payments after exactly 12 payments have been made. Furthermore, half of the consumers who have not defaulted after 12 payments default on all their remaining payments after exactly 18 payments have been made”. The bank also argues that “it costs 30% of each monthly repayment to collect the payment. These costs are incurred even if the payment is not made by the consumer. Furthermore, with inflation of 2.5% per annum, the real rate of interest that the lender obtains on the loan is less than 1.463% per annum effective”.

- i) Calculate the flat rate of interest paid by the consumer on the loan described above. (4)
 - ii) State why the flat rate of interest is not a good measure of the cost for borrowing to the consumer. (1)
 - iii) Using suitable supporting calculations, determine who is correct - the consumer forum or bank (or both?) (10)
- [15]**

Q. 7) A life insurance company issues a 3-year unit-linked endowment assurance contract to a male life aged 45 years exact with features as below:

Product features:

- Level premium of Rs. 20,000 per annum are payable in advance throughout the term of the policy or until earlier death.
- Death benefit is the higher of 3 times of level premium or bid value of the units and payable at the end of the policy year of death.
- There is nil surrender value at all times.
- On maturity, bid value of the units plus total policy fees collected during the entire policy term are payable.
- The premium allocated to units for 3 policy years is mentioned below:
 - 95% in the first year
 - 100% in the second year
 - 102.5% in the third year
- A policy fee of Rs. 150 is deducted from the bid value of units at the start of each policy year and inflates at the rate of 5% per annum from second year onwards.
- The units are subject to a bid-offer spread of 5%.
- An annual fund management charge of 1% of the bid value of the units is deducted at the end of each year. These management charges are deducted from the unit fund before death or maturity benefits are paid out.

Basis

The company uses the following assumptions for this contract for profit testing:

- Mortality: AM92 Ultimate
- Surrenders: None
- Initial Expense: Rs. 200 per policy
- Renewal Expenses: Rs. 100 per policy at the start of second year and inflating at the rate of 5% per annum for third policy year

- Initial Commission: 11% of first premium
 - Renewal Commission: 3% of second and third premium
 - Unit-fund growth rate: 8% per annum
 - Non-unit fund growth rate: 6% per annum
- i) Calculate the unit fund and non-unit fund at the end of each of the three policy years showing all components. Develop all cashflows rounded to nearest 2 decimal places. (15)
- ii) Without performing any additional calculations, explain whether this product causes any new business strain or not. (2)
- iii) Define Risk Discount Rate. Within a pricing basis, how would the investment return assumption and risk discount rate typically compare with each other? (3)
- iv) Explain what you understand by “bid-offer spread”? (2)
- v) Without performing any additional calculations, explain how the profitability of the product will be affected if maturity benefit is defined to be the bid value of units. (1)
- [23]**

Q. 8) A life insurance company issues a joint life annuity to a male life aged 55 years exact and a female life aged 50 years exact with following benefits:

- Lump sum of Rs. 1,00,000 is payable if both lives survive till the 5th policy anniversary.
- Annuity benefit payable monthly in advance with first payment to be made 5 years from the date of issue. The amount of annuity will be Rs. 20,000 per annum for as long as both lives are still alive or Rs. 10,000 while only one of them is alive.
- Lump sum of Rs. 1,00,000 is payable if either life survives till the 10th anniversary.

i) Calculate the present value of above benefits assuming the basis below:

Mortality: PMA92C20 and PFA92C20

Interest: 4% per annum

Expenses: Nil

(10)

ii) The insurance company is looking to invest the premiums from the products above in an asset class which would result in the closest effective duration matching. The company can invest in one of these three securities only: a 5-year zero coupon bond, 7-year zero coupon bond or 10-year zero coupon bond. Explain which of these securities should the insurance company select?

(3)
[13]

Q. 9) Two bonds paying annual coupons of 5% in arrear and redeemable at par have terms to maturity of exactly one year and two years, respectively. The gross redemption yield from the 1-year bond is 4.5% per annum effective; the gross redemption yield from the 2-year bond is 5.3% per annum effective. You are also informed that the 3-year par yield is 5.6% per annum.

Calculate all zero-coupon yields and all one-year forward rates implied by the yields given above.

[9]

- Q. 10)** A life assurance company issues a deferred post-retirement regular income plan to a life now aged 35 years exact. The policy provides an income of Rs. 50,000 per annum in advance starting on the policyholder's 60th birthday. The income increases by Rs 3,000 each year starting from the 61st birthday.

The policy also provides death benefit, payable immediately on death, which is given by the formula:

$10,000 \times (25 - t)$ where $t = 0, 1, 2, \dots, 24$ and t denotes the curtate duration in years since inception of the policy. Death benefit cover ceases at age 60 (i.e., at the end of the deferment period).

The policy is paid for by level monthly premiums payable in advance from the date of issue over the deferment period or until death if happens earlier.

The company uses the following premium basis for the policy:

- Mortality AM92 Select
- Interest 4% p.a.
- Initial commission 30% of the total premiums payable in the first policy year
- Initial expenses Rs.235
- Renewal commission 5% of the 2nd premium and subsequent monthly premiums
- Renewal expense Rs.60 p.a. at the start of the 2nd and subsequent policy years

The renewal expense is assumed to increase continuously at 4% per annum compound rate from inception of the policy. These expenses are incurred over the deferment period or until death if earlier.

Calculate monthly premium for the policy, show all your workings.

[14]
